

Conjugate Thermal Management of a Liquid-Cooled 14S4P Battery Pack: A COMSOL Multiphysics Study

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Table 1. Pack and simulation parameters.

Quantity	Value
Pack capacity (gross)	~ 120 kWh
Pack voltage (nominal)	~ 630 V (800 V class)
Cell chemistry	lithium manganese nickel
Simulated unit	14S4P, 8 modules
Cells per module	56
Cooling	cold plate + pipe network

Abstract

The life and fast-charge capability of a high-power battery pack are limited not only by peak temperature but by temperature uniformity between cells, since a spread drives uneven ageing and eventual imbalance. This report presents a coupled electrochemical–thermal COMSOL Multiphysics model of a liquid-cooled 14S4P subset of the Pininfarina Battista pack, in which the ohmic, activation and concentration losses of each cell enter the energy equation as heat sources and the cooling circuit is resolved by coupling non-isothermal turbulent flow in the cold plate to a non-isothermal pipe-flow network. Over a transient discharge cycle the design holds the temperature rise below about 8°C with no hotspots, keeps the cell voltage and state of charge uniform across modules, and limits state-of-health decay, which follows a temperature-dependent Arrhenius law, to above 0.80.

Keywords: battery thermal management; conjugate heat transfer; electrochemistry; COMSOL; liquid cooling; state of health

I Introduction

The usable energy, the fast-charge capability and the calendar life of a high-power battery pack are all governed by temperature, and not only by its peak value. What matters as much is uniformity: a temperature spread between cells makes some cells age faster than others, and that divergence eventually shows up as capacity imbalance and a shortened pack life. A thermal-management system therefore has two jobs, to remove heat and to keep the cells at a common temperature under load.

This report studies a plate-and-pipe liquid-cooling design for a subset of the Pininfarina Battista pack, a roughly 120 kWh, ~ 630 V automotive pack (Table 1), using a coupled electrochemical–thermal model built in COMSOL Multiphysics. The aim is to verify that the cooling design holds cell-to-cell uniformity through a demanding discharge cycle.

II Model

Each cell carries an electrochemical model whose irreversible losses enter the energy balance as volumetric heat sources,

$$\mathcal{Q} = \mathcal{Q}_{\text{ohm}} + \mathcal{Q}_{\text{act}} + \mathcal{Q}_{\text{conc}}, \quad (1)$$

the ohmic, activation and concentration overpotentials respectively. The temperature field obeys the convection–diffusion energy equation

$$\rho C_p \left(\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) = \nabla \cdot (k \nabla T) + \mathcal{Q}, \quad (2)$$

with $\mathbf{u}$ the coolant velocity and k the thermal conductivity. The cooling circuit is resolved by coupling a non-isothermal turbulent flow in the cold plate to a non-isothermal pipe-flow network, so that the conjugate heat transfer between cells, plate and coolant is captured directly. Cell ageing is tracked through a state-of-health variable governed by a temperature-dependent Arrhenius law,

$$\frac{d(\text{SOH})}{dt} \propto -A \exp\left(-\frac{E_a}{RT}\right), \quad (3)$$

so that warmer cells degrade faster through accelerated solid-electrolyte-interphase growth and side reactions. The local electric potential through the pack (Fig. 2) and the distribution of heat sources (Fig. 3) close the coupling between the electrochemical and thermal fields.

III Results

A transient discharge cycle was simulated over the module set. The pack temperature field (Fig. 1) stays within a modest band, with a peak cell temperature of about 27.6°C and a temperature rise held below roughly 8°C despite continuous discharge, and no hotspots, a direct consequence of the optimized pipe layout. The cell voltage and applied current (Fig. 4) remain stable through most of the cycle, dropping only near the end as the cells deplete normally, with improved voltage stability at the slightly elevated operating temperature. The state of charge falls uniformly across the three probed cells (Fig. 5), confirming balanced consumption and the absence of current imbalance. The modelled state of health (Fig. 6) stays above 0.80 over the cycle; it decays slowly at first, to about 0.99 by 600 s, and then accelerates with temperature and overpotential, exactly as the Arrhenius model predicts. The homogeneous degradation of the probed cells confirms that the cooling plate and flow distribution impose no significant thermal or current imbalance.

IV Conclusion

The coupled electrochemical–thermal model shows that a plate-and-pipe liquid-cooling design holds the Battista-class

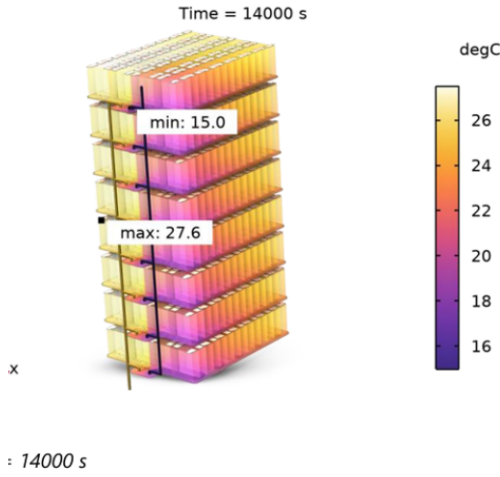


Figure 1. Three-dimensional pack temperature field at the end of the discharge cycle (minimum 15.0, maximum 27.6 °C).

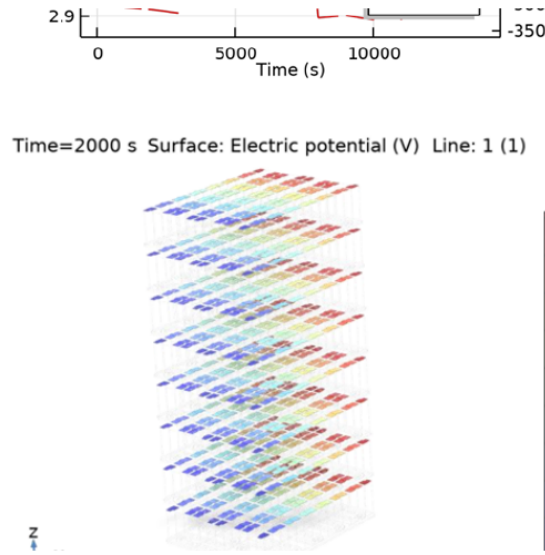


Figure 2. Local electric potential through the pack, coupling the electrochemical and thermal fields.

pack to a small, uniform temperature rise under high discharge, which protects both safety and even ageing. The same model is directly reusable as a design tool: the pipe layout, coolant flow rate and cell spacing can be swept against the temperature spread and the state-of-health decay to size a thermal system for a target duty cycle.

References

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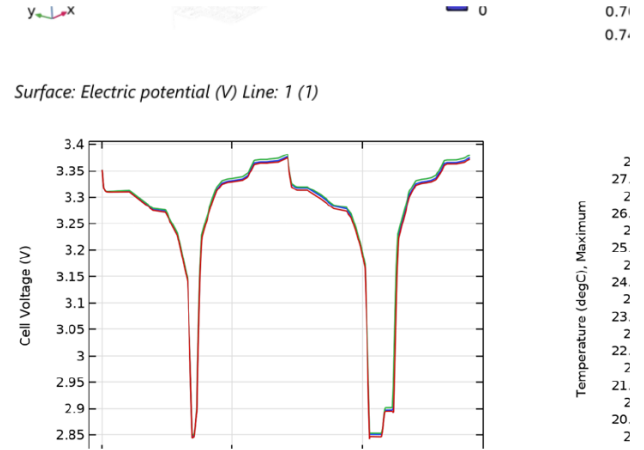


Figure 3. Distribution of volumetric heat sources across the module set.

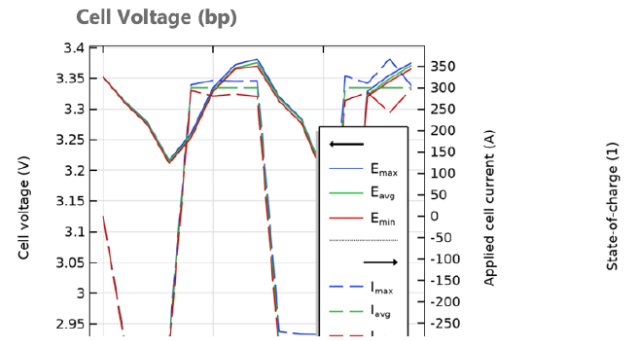


Figure 4. Cell voltage and applied cell current through the discharge cycle.

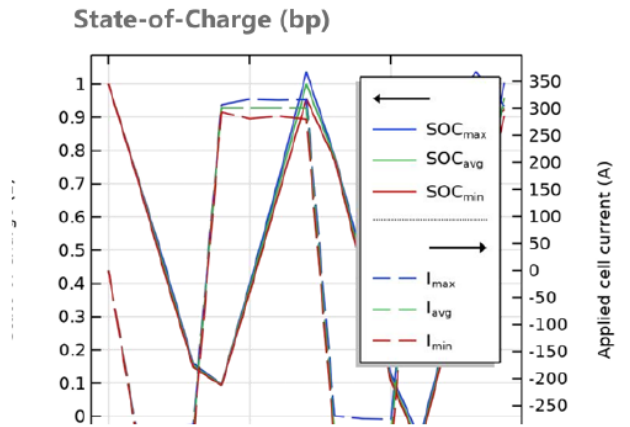


Figure 5. State of charge of the probed cells, falling uniformly across the modules.

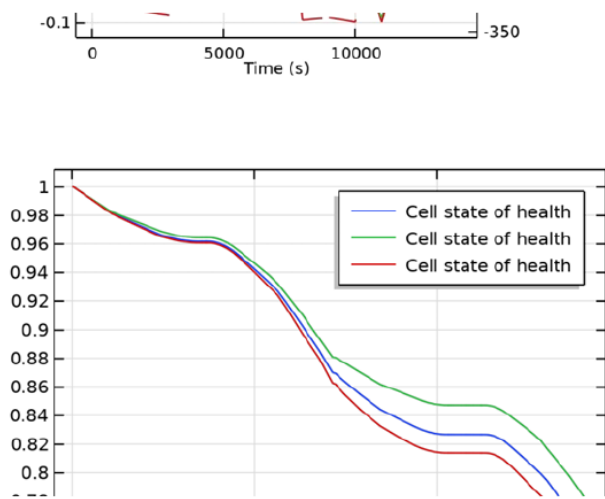


Figure 6. Modelled state-of-health decay, correlated with local cell temperature.